Arsh Satsangi

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Dayalbagh Educational Institute
Dayalbagh, Agra, U.P. 282110

Summary

Electrical Engineering undergraduate (Electronics Specialization) at DEI, Agra, skilled in embedded systems, PCB design, analog circuits, and FPGA/microcontroller prototyping. Backed by industrial transformer testing experience at MEI Power and top global hackathon rankings, seeking an engineering internship to bridge circuit simulation and hardware implementation.

Education

Qualification	Institution	Marks	Year
B.Tech, Electrical Engineering	Dayalbagh Educational Institute (DEI), Agra	7.07 CGPA	2028
Class XII	Radhasoami Educational Institute (REI) Intermediate College, Agra	8.00 CGPA	2024
Class X	Radhasoami Educational Institute (REI) Intermediate College, Agra	8.60 CGPA	2022

Experience

Electrical Engineering Intern (one-month training) May–June, 2026
MEI Power Private Limited (MARSONS Group), Agra, Uttar Pradesh

- Assisted in design, assembly, and testing of heavy-duty power and distribution transformers, verifying electrical parameters against Indian Standards (IS) and International Electrotechnical Commission (IEC) quality and energy-efficiency standards.
- Gained hands-on exposure to core lamination, winding techniques, and insulation testing under senior engineers.

Projects

VLSI Design of a 32-bit CORDIC Processor for Trigonometric Calculation Verilog | Xilinx Vivado | Fixed-Point Arithmetic

- Designed a 32-bit Coordinate Rotation Digital Computer (CORDIC) processor in Verilog that computes sine and cosine using only shifts, additions, and subtractions, with no hardware multipliers; a 4-bit synchronous counter and a 16-entry arctangent Look-Up Table (LUT) replace a complex Finite State Machine (FSM), giving 16 clock cycles per result.
- Built a quadrant-folding wrapper (`cordic_sincos.v`) for angles from $-\pi$ to $+\pi$ in Q3.29 format, and a 16-stage pipelined core (`cordic_pipelined.v`) that delivers one result per clock cycle once the pipeline fills.
- Used 32-bit signed Q2.30 fixed-point arithmetic (resolution of about 9.31×10^{-10}) with a worst-case absolute error of about 3.01×10^{-5}, roughly 15 bits of precision.
- Verified in Xilinx Vivado using self-checking Verilog testbenches cross-checked against a bit-accurate Python reference model; placed-and-routed layouts visualised.

WireGuard: Smart Detection of Power Line Failures Analog Circuits | Signal Filtering | Resonance Tuning

- Designed a low-cost analog fault-detection system for low-voltage distribution lines, removing complex digital infrastructure while preserving real-world applicability.
- Tuned distribution poles to unique resonant frequencies to enable precise identification of the exact line-break location.
- Validated through field testing: achieved a 50-metre communication range along the power line with consistent reliability.
- Mitigated harmonics and Total Harmonic Distortion (THD) via signal filtering, frequency isolation, and resonance stability techniques.

Technical Skills

Languages	C, Verilog
Embedded and Hardware	Embedded Systems, Microcontroller Programming, Raspberry Pi
PCB and EDA Tools	Altium Designer, KiCad, Xilinx Vivado, LTspice, AutoCAD
Core Competencies	PCB Design, Digital Logic Design, Analog Circuit Analysis,
Test and Instruments	Oscilloscopes, Multimeters, Function Generators, Soldering and Rework
Software and OS	Linux, GitHub,

Achievements

- HackSecure X International Hackathon, 2026: Top 11 among 250 teams globally in a cybersecurity-focused hardware challenge.
- DEI Virtual Labs Hackathon, 2026: Top 20 among 200 teams at the national level.
- Anveshan Zonal Research Convention, 2025: Qualified for the North Zone Student Research Convention, representing DEI.
- Cybersecurity Awareness Quiz, Institute of Electrical and Electronics Engineers (IEEE) DEI: 1st position among all participants.

Personal Details

Date of birth	07 May 2007	Languages known	English,
Father's name	Deepak Kumar	Interests	Hardware Prototyping, Circuit Design,
Address	Dayalbagh, Agra, Uttar Pradesh, 282110	Hobbies	Football, Badminton

I hereby declare that the information furnished above is true and correct to the best of my knowledge. Arsh Satsangi